Cainiao is Fast — But Not Everywhere

When does Alibaba's logistics network stop delivering its speed advantage?

Group 12

BUAD 345: Decision Analytics and Visualization

Summer 2026

The question

- **Audience: Cainiao** (Alibaba's logistics platform).
- Delivery speed drives sales and loyalty: online sales rise ∼1.45% per business-day faster delivery (Fisher et al., 2019); late delivery erodes loyal customers (Rao et al., 2011).
- Cainiao handles **30.2%** of orders — does it actually deliver faster, and *everywhere*?

Data: 1.29M Tmall orders, Jan–Jul 2017 (order + 969 MB logistics + merchant files).

Method: dig hierarchically, not in parallel

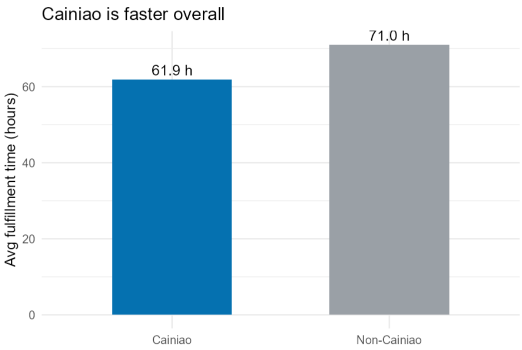
$$\text{totaltime} = \text{SIGNED} - \text{pay\_timestamp (hours)}$$

1. Cainiao vs. non-Cainiao
2. + carrier size (Large >200k orders)
3. + destination-city size (top-2 signing cities = Big)
4. segment decomposition: pre-delivery / line-haul / last-mile

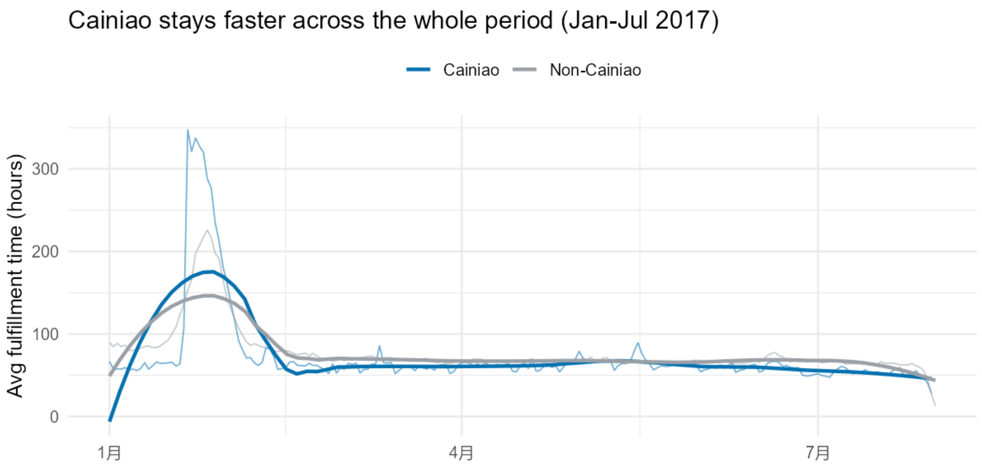
One focused theme; each level added only when the previous result motivates it.

Headline: Cainiao is faster overall

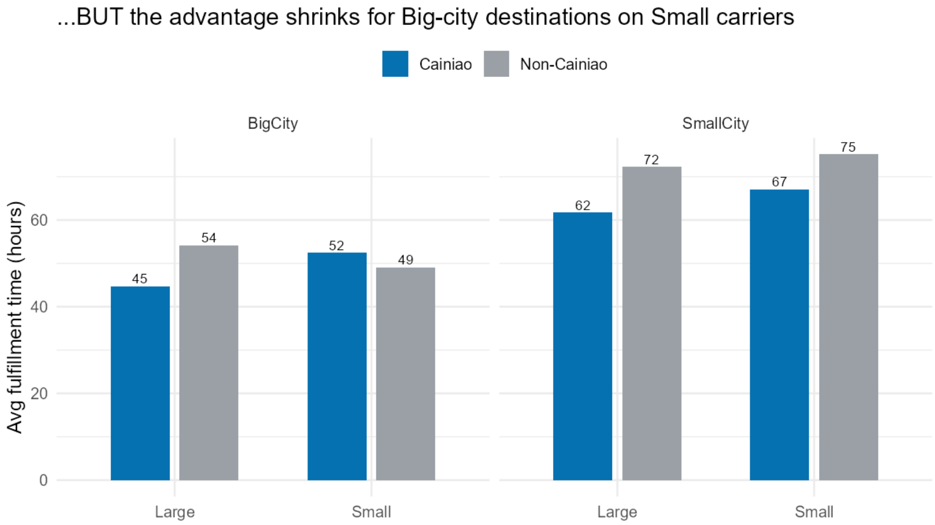
- **61.9 h** vs. **71.0 h** $\Rightarrow$ $\sim$9.2 h faster.
- Survives a fair control: among no-promise orders, 62.5 vs. 71.0 h.
- Stable across the whole 7-month window.



Stable over time



...BUT the advantage is uneven — the punchline



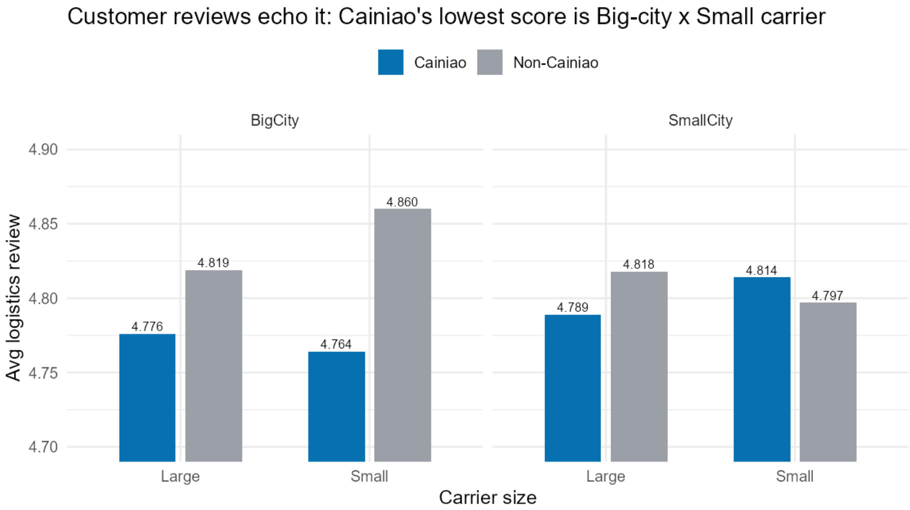
For **big-city destinations on small carriers**, Cainiao is 3.3 h *slower* than non-Cainiao.

The reversal cell, in numbers

Destination	Carrier	Cainiao	Non-Cainiao	**Cainiao saved**
Big city	Large	44.7	54.2	**+9.5 h**
Big city	Small	52.4	49.1	**−3.3 h**
Small city	Large	61.7	72.2	**+10.5 h**
Small city	Small	67.1	75.2	**+8.0 h**

22,756 orders in the reversal cell — not a small-sample fluke.

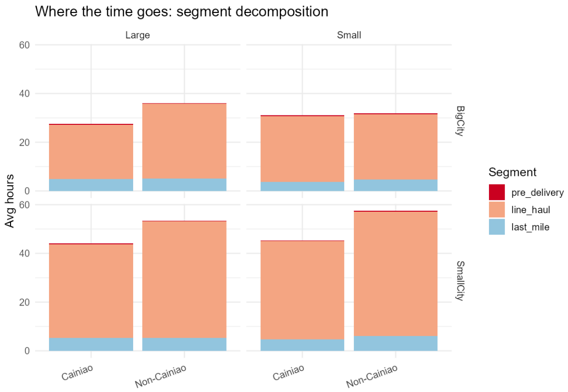
Customers feel it too



Cainiao's *lowest* review (4.764) is the same big-city / small-carrier cell; non-Cainiao's *highest* (4.860) is right beside it

Mechanism: it's line-haul, not the last mile

- Last-mile is small & similar everywhere ($\sim$4–6 h).
- Cainiao's edge = faster **line-haul** (e.g. 38.4 vs. 47.8 h).
- In the reversal cell that edge vanishes (27.0 vs. 26.9 h).



An honest non-result

- Hypothesis: small carriers make more stops $\Rightarrow$ slower.
- **Not supported.** Small carriers pass through *fewer* cities (2.9–3.1) than large ones (3.4–3.7).
- So the penalty is *per-leg* transit/dwell time, not more hops — which sharpens the explanation we give.

Why

- Big cities concentrate trunk-line volume; large carriers run dedicated, high-frequency line-haul, small carriers wait to consolidate (Yu et al., 2016).
- Small carriers can't execute one-size-fits-all smart routing; ignoring carrier heterogeneity degrades routing performance (Koç et al., 2016).

So what — three moves for Cainiao

1. **Deepen ties with large carriers** on big-city lanes (watch concentration risk).
2. **Make smart-routing carrier-capability-aware** (size, coverage, experience).
3. **Pilot big-city consolidation / parcel lockers** for small-carrier shipments to cut the line-haul + pickup penalty (Deutsch & Golany, 2018).

The weakness is concentrated, so the fix is cheap vs. a platform-wide change.

Summary in two sentences

Overall

Cainiao significantly shortens fulfillment time across the marketplace ($\sim$9 h, robust to a promise control).

Punchline

But that advantage largely disappears — and reverses — for big-city destinations served by small carriers, with the bottleneck in line-haul rather than the last mile.

Limitations: observational data (association, not causation); missing milestones; coarse big-city proxy.

Thank you — Group 12

Questions?